Supplementary Materials

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Table S1. Summary of amendments to information provided at registration or in the protocol.

Information at registration or in protocol	Amendments
Mean change from baseline mean and standard deviation (SD) would be used as the measure of effect	Where mean change from baseline could not be imputed for a study, post-intervention mean and SD were used in the meta-analysis.
Sensitivity analysis did not include restricting the analysis to a subset of the studies with medications already approved by the government agency e.g., the United States Food and Drug Administration.	A sensitivity analysis was conducted to repeat the meta-analysis which included only a subset of the studies with medications already approved (excluding medications under development).
Publication bias would be assessed via visual inspection of funnel plots and the Egger's test	Clarification was added to specify that publication bias was assessed via visual inspection of funnel plots and, where there were ≥10 studies, the Egger's test.
Sensitivity and subgroup analysis would be performed to assess the robustness of results.	To help with the interpretations, a threshold of what would be considered a 'relevant change' to the results was introduced (Results are considered changed if the result direction changed between effect and no effect or if g-value changed ≥0·3)

Table S2. Search terms and database search results

From date of database inception to 10 July 2022

PubMed		
Search	**Query**	**Hits**
#1	"back pain"[MeSH Terms] OR "low back pain"[MeSH Terms] OR "back pain* "[Title/Abstract] OR "lumbago"[Title/Abstract] OR "backache*"[Title/Abstract] OR "back ache*"[Title/Abstract] OR "spinal stenosis"[Title/Abstract] OR "canal stenosis"[Title/Abstract] OR "lumbar stenosis"[Title/Abstract] OR "lateral stenosis "[Title/Abstract] OR "neurogenic claudication"[Title/Abstract] OR "radiculopathy"[Title/Abstract] OR "radicular pain"[Title/Abstract] OR "spondylolisthesis"[Title/Abstract] OR "spondylosis"[Title/Abstract] OR "sciatica"[Title/Abstract] OR "intervertebral disc displacement"[Title/Abstract] OR "spinal nerve roots"[Title/Abstract] OR "neurologic signs "[Title/Abstract] OR "paresthesia"[Title/Abstract] OR "paraesthesia"[Title/Abstract] OR "numbness"[Title/Abstract]	106522
#2	"sleep"[MeSH Terms] OR "sleep initiation and maintenance disorders"[MeSH Terms] OR "sleep*" OR "insomnia" OR "nap" OR "naps" OR "napping"	265333
#3	("randomized controlled trial"[Publication Type] OR "controlled clinical trial"[Publication Type] OR "randomized"[Title/Abstract] OR "placebo"[Title/Abstract] OR "randomly"[Title/Abstract] OR "trial"[Title] OR "clinical trials as topic"[MeSH Terms:noexp]) NOT (animals[MeSH Terms] NOT humans[MeSH Terms])	1272008
#4	animals[MeSH Terms] NOT humans[MeSH Terms]	4835945
#5	#1 AND #2 AND #3 NOT #4	358
CINAHL		
Search	**Query**	**Hits**
#1	MH "back pain+" OR MH "low back pain+" OR TI ("back pain*" OR "lumbago" OR "backache*" OR "back ache*" OR "spinal stenosis" OR "canal stenosis" OR "lumbar stenosis" OR "lateral stenosis " OR "neurogenic claudication" OR "radiculopathy" OR "radicular pain" OR "spondylolisthesis" OR "spondylosis" OR "sciatica" OR "intervertebral disc displacement" OR "spinal nerve roots" OR "neurologic signs " OR "paresthesia" OR "paraesthesia" OR "numbness") OR AB ("back pain*" OR "lumbago" OR "backache*" OR "back ache*" OR "spinal stenosis" OR "canal stenosis" OR "lumbar stenosis" OR "lateral stenosis " OR "neurogenic claudication" OR "radiculopathy" OR "radicular pain" OR "spondylolisthesis" OR "spondylosis" OR "sciatica" OR "intervertebral disc displacement" OR "spinal nerve roots" OR "neurologic signs " OR "paresthesia" OR "paraesthesia" OR "numbness")	51148
#2	MH "sleep+" OR MH "sleep initiation and maintenance disorders+" OR "sleep*" OR "insomnia" OR "nap" OR "naps" OR "napping"	87266
#3	PT("randomized controlled trial" OR "controlled clinical trial") OR TI("randomized" OR "placebo" OR "randomly" OR "trial") OR AB("randomized" OR "placebo" OR "randomly"	371753
#4	MH "animals+" NOT MH "human+"	87756
#5	#1 AND #2 AND #3 NOT #4	148
SPORTDiscus		
Search	**Query**	**Hits**
#1	DE "backache+" OR TI ("back pain*" OR "lumbago" OR "backache*" OR "back ache*" OR "spinal stenosis" OR "canal stenosis" OR "lumbar stenosis" OR "lateral stenosis " OR "neurogenic claudication" OR "radiculopathy" OR "radicular pain" OR "spondylolisthesis" OR "spondylosis" OR "sciatica" OR "intervertebral disc displacement" OR "spinal nerve roots" OR "neurologic signs " OR "paresthesia" OR "paraesthesia" OR "numbness") OR AB ("back pain*" OR "lumbago" OR "backache*" OR "back ache*" OR "spinal stenosis" OR "canal stenosis" OR "lumbar stenosis" OR "lateral stenosis " OR "neurogenic claudication" OR "radiculopathy" OR "radicular pain" OR "spondylolisthesis" OR "spondylosis" OR "sciatica" OR "intervertebral disc displacement" OR "spinal nerve roots" OR "neurologic signs " OR "paresthesia" OR "paraesthesia" OR "numbness")	11290
#2	DE "sleep+" OR "sleep*" OR "insomnia" OR "nap" OR "naps" OR "napping"	14747

#3	TI("randomized" OR "placebo" OR "randomly" OR "trial") OR AB("randomized" OR "placebo" OR "randomly")	54327
#4	#1 AND #2 AND #3	30

PsychINFO

#1	MA "back pain+" OR MA "low back pain+" OR TI ("back pain*" OR "lumbago" OR "backache*" OR "back ache*" OR "spinal stenosis" OR "canal stenosis" OR "lumbar stenosis" OR "lateral stenosis " OR "neurogenic claudication" OR "radiculopathy" OR "radicular pain" OR "spondylolisthesis" OR "spondylosis" OR "sciatica" OR "intervertebral disc displacement" OR "spinal nerve roots" OR "neurologic signs " OR "paresthesia" OR "paraesthesia" OR "numbness") OR AB ("back pain*" OR "lumbago" OR "backache*" OR "back ache*" OR "spinal stenosis" OR "canal stenosis" OR "lumbar stenosis" OR "lateral stenosis " OR "neurogenic claudication" OR "radiculopathy" OR "radicular pain" OR "spondylolisthesis" OR "spondylosis" OR "sciatica" OR "intervertebral disc displacement" OR "spinal nerve roots" OR "neurologic signs " OR "paresthesia" OR "paraesthesia" OR "numbness")	8007
#2	MH "sleep+" OR MH "sleep initiation and maintenance disorders+" OR "sleep*" OR "insomnia" OR "nap" OR "naps" OR "napping"	81537
#3	TI("randomized" OR "placebo" OR "randomly" OR "trial") OR AB("randomized" OR "placebo" OR "randomly")	179418
#4	MA "animals+" NOT MA "human+"	42284
#5	#1 AND #2 AND #3 NOT #4	56

EMBASE

Search	Query	Hits
#1	'back pain'/exp OR 'low back pain'/exp OR 'back pain*':ab,ti OR 'lumbago':ab,ti OR 'backache*':ab,ti OR 'back ache*':ab,ti OR 'spinal stenosis':ab,ti OR 'canal stenosis':ab,ti OR 'lumbar stenosis':ab,ti OR 'lateral stenosis':ab,ti OR 'neurogenic claudication':ab,ti OR 'radiculopathy':ab,ti OR 'radicular pain':ab,ti OR 'spondylolisthesis':ab,ti OR 'spondylosis':ab,ti OR 'sciatica':ab,ti OR 'intervertebral disc displacement':ab,ti OR 'spinal nerve roots':ab,ti OR 'neurologic signs':ab,ti OR 'paresthesia':ab,ti OR 'paraesthesia':ab,ti OR 'numbness':ab,ti	184874
#2	'sleep'/exp OR 'sleep*' OR 'insomnia' OR 'nap' OR 'naps' OR 'napping'	492802
#3	'randomized controlled trial':it OR 'controlled clinical trial':it OR 'randomized':ab,ti OR 'placebo':ab,ti OR 'randomly':ab,ti OR 'trial':ti OR 'clinical trials'/de	1562710
#4	'animal'/exp NOT 'human'/exp	5616914
#5	#1 AND #2 AND #3 NOT #4	2836

CENTRAL

Search	Query	Hits
#1	(MeSH descriptor: [Low Back Pain] explode all trees) OR (MeSH descriptor: [Back Pain] explode all trees) OR ("back pain*" OR "lumbago" OR "backache*" OR "back ache*" OR "spinal stenosis" OR "canal stenosis" OR "lumbar stenosis" OR "lateral stenosis " OR "neurogenic claudication" OR "radiculopathy" OR "radicular pain" OR "spondylolisthesis" OR "spondylosis" OR "sciatica" OR "intervertebral disc displacement" OR "spinal nerve roots" OR "neurologic signs " OR "paresthesia" OR "paraesthesia" OR "numbness"):ti,ab,kw	24838
#2	(MeSH descriptor: [Sleep] explode all trees) OR (MeSH descriptor: [Sleep Initiation and Maintenance Disorders] explode all trees) OR ("sleep*" OR "insomnia" OR "nap" OR "naps" OR "napping")	47663
#3	(MeSH descriptor: [Animals] explode all trees) NOT (MeSH descriptor: [Humans] explode all trees)	59
#4	#1 AND #2 NOT #3	1504
#5	#4 AND Limit: Trials	1462

Table S3. GRADE criteria

GRADE Domain	Criteria used in this review
Risk of bias	Downgraded by one level if 50% of studies or 50% of n-randomised had high risk of bias, or 75% of studies or 75% of n-randomised was from studies with some concerns. Downgraded by two levels if all studies had high risk of bias
Inconsistency	Downgraded by one level if substantial heterogeneity was present. For example, effect estimates varied widely across studies, 95% CI had showed minimal or no overlap, I^2>50%, or the prediction interval showing treatment effects favouring both directions.
Indirectness	Downgraded by one level if any evidence supporting the meta-analysis came from studies not directly related to the Population, Intervention, Comparators, Outcomes, Study type (PICOS) framework.
Imprecision	Downgraded by one level if the 95% confidence interval of the effect estimate includes no effect or the sample size <800 in total (<400 per group).
Publication bias	Downgraded by one level if evidence of asymmetry in the funnel plots, or statistical evidence (p<·05) from the modified Eggers test. Downgraded by one level if ≥75% of studies were industry funded.

Table S4. A selected list of excluded studies with reasons

This was not a comprehensive list of excluded studies. Rather, this is a selected list to demonstrate some key exclusion.

Title	Year	Author	Reason for exclusion
A randomized, double-blind, placebo-controlled trial of eszopiclone for the treatment of insomnia in patients with chronic low back pain.	2014	Goforth HW, Preud'homme XA, Krystal AD	Ineligible comparator. The comparator included a therapeutic component.
A randomized, open-label study of once-a-day AVINZA®(morphine sulfate extended-release capsules) versus twice-a-day OxyContin®(oxycodone hydrochloride controlled-release tablets) for chronic low back pain: The extension phase of the ACTION trial	2006	Rauck RL, Bookbinder SA, Bunker TR, et al.	Ineligible study design. The study was a randomised withdrawal trial.
A subgroup analysis found no diminished response to buprenorphine transdermal system treatment for chronic low back pain patients classified with depression.	2016	Yarlas A, Miller K, Wen W, et al.	Ineligible study design. The study was a randomised withdrawal trial.
Analgesic efficacy and safety of oxycodone in combination with naloxone as prolonged release tablets in patients with moderate to severe chronic pain	2008	Vondrackova D, Leyendecker P, Meissner W, et al.	Ineligible study design. The study was a randomised withdrawal trial.
Analgesic efficacy and tolerability of tramadol 100 mg sustained-release capsules in patients with moderate to severe chronic low back pain	1999	Raber M, Hofmann S, Junge K, Momberger H, Kuhn D	Ineligible comparator. The comparator included a therapeutic component.
Botulinum toxin treatment for nocturnal calf cramps in patients with lumbar spinal stenosis: a randomized clinical trial.	2017	Park SJ, Yoon KB, Yoon DM, Kim SH	Ineligible comparator. The comparator included a therapeutic component.
Buprenorphine transdermal system in adults with chronic low back pain: a randomized, double-blind, placebo-controlled crossover study, followed by an open-label extension phase.	2010	Gordon A, Callaghan D, Spink D, et al.	Ineligible comparator. The comparator included a therapeutic component.
Clinical Effects and Safety of the Use of Methylene Blue for the Treatment of Lumbar Facet Joint Syndrome	2022	Yu X, Zhang J, Wang M, et al.	Ineligible comparator. The comparator included a therapeutic component.
Clinical effect modifiers of antibiotic treatment in patients with chronic low back pain and Modic changes - secondary analyses of a randomised, placebo-controlled trial (the AIM study).	2020	Bråten LCH, Grøvle L, Espeland A, et al.	Ineligible outcome. No sleep outcome was reported.

Title	Year	Author	Reason for exclusion
Combination therapy with transdermal buprenorphine and pregabalin for chronic low back pain.	2012	Pota V, Barbarisi M, Sansone P, et al.	Ineligible study design. The study was a randomised withdrawal trial.
Comparative effects of morning vs. evening dosing of extended-release hydromorphone on sleep physiology in patients with low back pain: a pilot study.	2015	Webster LR, Smith MD, Mackin S, Iverson M	Ineligible comparator. The comparator included a therapeutic component.
Comparison of the efficacy and tolerability of a paracetamol/codeine fixed-dose combination with tramadol in patients with refractory chronic back pain.	1998	Müller FO, Odendaal CL, Müller FR, Raubenheimer J, Middle MV, Kummer M	Ineligible comparator. The comparator included a therapeutic component.
Controlled-release oxycodone and naloxone in the treatment of chronic low back pain: a placebo-controlled, randomized study	2013	Cloutier C, Taliano J, O'Mahony W, et al.	Ineligible comparator. The comparator included a therapeutic component.
Doxepin as an adjunct in the treatment of chronic pain	1982	Aronoff GM, Evans WO	Ineligible study design. The study was a review.
Doxepin's effects on chronic pain and depression: A controlled study	1984	Hameroff SR, et al.	Ineligible participants (mixed participants). The study included participants with ‘chronic cervical and/or lumbar spine pain' without reporting separate data for participants with chronic lumbar spine pain.
Effect of opioid vs nonopioid medications on pain-related function in patients with chronic back pain or hip or knee osteoarthritis pain: the SPACE randomized clinical trial	2018	Krebs EE, Gravely A, Nugent S, et al.	Ineligible comparator. The comparator included a therapeutic component.
Effectiveness and tolerability of a moderate dose of tapentadol prolonged release for managing severe, chronic low back pain with a neuropathic component: an open-label continuation arm of a randomized phase 3b study	2015	Baron R, Kern U, Müller M, DuboisC, FalkeD, Steigerwald I	Ineligible comparator. The comparator included a therapeutic component.
Efficacy and safety of a hydrocodone extended-release tablet formulated with abuse-deterrence technology in patients with moderate-to-severe chronic low back pain	2015	Hale ME, Zimmerman TR, Eyal E, Malamut, R.	Ineligible study design. The study was a randomised withdrawal trial.
efficacy and safety of buprenorphine transdermal system (BTDS) for chronic moderate to severe low back pain: a randomized, double-blind study	2011	Steiner D, Munera C, Hale M, Ripa S, Landau C.	Ineligible comparator. The comparator included a therapeutic component.
Efficacy and safety of controlled-release oxycodone for the management of moderate-to-severe chronic low back pain in	2019	Kawamata M, Iseki M, Kawakami M, et al.	Ineligible study design. The study was a randomised withdrawal trial.

Title	Year	Author	Reason for exclusion
Japan: results of an enriched enrollment randomized withdrawal study followed by an open-label extension study			
Efficacy and safety of oxymorphone extended release in chronic low back pain: Results of a randomized, double-blind, placebo- and active-controlled phase III study	2005	Hale ME, Dvergsten C, Gimbel J	Ineligible study design. The study was a randomised withdrawal trial.
Efficacy and safety of the seven-day buprenorphine transdermal system in opioid-naïve patients with moderate to severe chronic low back pain: an enriched, randomized, double-blind, placebo-controlled study.	2011	Steiner DJ, Sitar S, Wen W, et al.	Ineligible study design. The study was a randomised withdrawal trial.
Efficacy and tolerability of buccal buprenorphine in opioid-naive patients with moderate to severe chronic low back pain	2016	Rauck RL, Potts J, Xiang Q, Tzanis E, Finn A	Ineligible study design. The study was a randomised withdrawal trial.
Efficacy and tolerability of sustained-release ibuprofen in the treatment of patients with chronic back pain	1994	Driessens M, Famaey JP, Orloff S, et al.	Ineligible comparator. The comparator included a therapeutic component.
Efficacy, safety, and tolerability of nktr-181 in patients with moderate to severe chronic low-back pain: A phase 3 study	2017	Markman J, Gudin J, Rauck R, Argoff C, Rowbotham M, Agaiby E, et al.	Ineligible record type. The record was grey literature (abstract).
Extended-release gabapentin for failed back surgery syndrome: Results from a randomized double-blind cross-over study	2019	Gewandter JS, Frazer ME, Cai X, et al.	Ineligible outcome. Sleep outcome was mentioned but no sleep data was reported.
Extended-release tramadol (tramadol ER) in the treatment of chronic low back pain.	2008	Vorsanger GJ, Xiang J, Gana TJ, Pascual ML, Fleming RR	Ineligible study design. The study was a randomised withdrawal trial.
Intrathecal midazolam for the treatment of chronic mechanical low back pain: a controlled comparison with epidural steroid in a pilot study.	1992	Serrao JM, Marks RL, Morley SJ, Goodchild CS	Ineligible comparator. The comparator included a therapeutic component.
Opioid therapy for chronic noncancer back pain. A randomized prospective study.	1998	Jamison RN, Raymond SA, Slawsby EA, Nedeljkovic SS, Katz NP	Ineligible comparator. The comparator included a therapeutic component.
Response to duloxetine in chronic low back pain: exploratory post hoc analysis of a Japanese phase III randomized study	2017	Tsuji T, Itoh N, Ishida M, Ochiai T, Konno S	Ineligible outcome. The study measured and reported 'pain interference with sleep' rather than a sleep measure.
Sleep disturbance predicts less improvement in pain outcomes: secondary analysis of the SPACE randomized clinical trial.	2020	Koffel E, Kats AM, Kroenke K, et al.	Ineligible comparator. The comparator included a therapeutic component.
SUMMIT-07: a randomized trial of NKTR-181, a new molecular entity, full mu-opioid receptor agonist for chronic low-back pain	2019	Markman J, Gudin J, Rauck R, et al.	Ineligible study design. The study was a randomised withdrawal trial.

Title	Year	Author	Reason for exclusion
The ACTION study: a randomized, open-label, multicenter trial comparing once-a-day extended-release morphine sulfate capsules (AVINZA) to twice-a-day controlled-release oxycodone hydrochloride tablets (OxyContin) for the treatment of chronic, moderate to	2006	Rauck RL, Bookbinder SA, Bunker TR, et al.	Ineligible comparator. The comparator included a therapeutic component.
The clinical difference between gabapentin and pregabalin: data from a pilot comparative trial.	2021	Gammoh O, Al-Smadi A, Shawagfeh MQ, et al.	Ineligible comparator. The comparator included a therapeutic component.
The effectiveness of gabapentin in patients with chronic radiculopathy	2003	Yildirim K, Şışecıoğlu M, Karatay S, et al.	Ineligible outcome. No sleep outcome was reported.
The efficacy and safety of pregabalin in the treatment of neuropathic pain associated with chronic lumbosacral radiculopathy	2010	Baron R, Freynhagen R, Tölle TR, et al.	Ineligible study design. The study was a randomised withdrawal trial.
The role of melatonin in the treatment of chronic back pain	2016	Kurganova YM, Danilov AB	Ineligible comparator. The comparator included a therapeutic component.
Tricyclic antidepressants for chronic low-back pain. Mechanisms of action and predictors of response	1986	Ward NG	Ineligible outcome (no sleep outcomes).

Table S5. Data handling

Authors have been contacted for missing data. Where responses were not received or insufficient data remained available, we handled the data as per the process below. Cochrane formulae were used when imputing SD from standard error (SE), p-values, or confidence intervals.[a]

Study	Outcome	Data handling
Buynak 2010[60]	[S] Sleep questionnaire	Insufficient data to allow pooling. The study was omitted from all meta-analyses.
Christoph 2017[57]	Overall	To prevent the unit of analysis error, data from cebranopadol 200/400/600ug were pooled into one treatment arm and n in the control group were proportionally divided for each pairwise meta-analysis.
	[S] Chronic Pain Sleep Inventory-sleep quality	Reverse-scaled means were multiplied by -1 to ensure the higher scores reflected worse health outcomes.
	[P] Numeric rating scale	SE was converted to SD
Cloutier 2013[58]	Overall	As n analysed for each crossover arm was not reported, n was proportionally divided to prevent unit of analysis errors.
	[S] Pain and Sleep Questionnaire	- SD for mean change from baseline was calculated using p-value for mean change data for each group. - Reverse-scaled means were multiplied by -1
Gordon 2010a[61]	Overall	As n analysed for each crossover arm was not reported, n was proportionally divided to prevent unit of analysis errors.
	[S] Pain and Sleep Questionnaire	- SD for mean change from baseline was calculated using p-value for mean change data for each group. - Data for each sleep characteristic were extracted from images - SD for 'trouble falling asleep' were imputed from pooled SD from Cloutier 2013 and Gordon 2010b
	[P] VAS [D] Quebec Back Pain Disability Scale	- SD for mean change from baseline was calculated using p-value for mean change data for each group.
Gordon 2010b[62]	Overall	As n analysed for each crossover arm was not reported, n was proportionally divided to prevent unit of analysis errors.
	[S] Pain and Sleep Questionnaire	- SD for mean change from baseline was calculated using p-value for mean change data for each group. - Reverse-scaled means were multiplied by -1
	[P] Visual analogue scale	SD for mean change from baseline was calculated using p-value for mean change data for each group.
Lee 2013[59]	[S],[P],[D] Oswestry Disability Index	Mean change was multiplied by -1 as the study reported improvement in the health outcome rather than worse.
Manning 2017[56]	[S] Numeric rating scale	- SE was converted to SD - Data from 'trouble falling asleep' and 'trouble staying asleep' were pooled to provide a global score for the 'Sleep (overall)' outcome

Study	Outcome	Data handling
Skljarevski 2009[55]	Overall	Data from duloxetine 20/60/120 mg were pooled into one treatment arm and n in the control group were proportionally divided for each pairwise meta-analysis.
	[S] Athens Insomnia Scale [P] Brief Pain Inventory-average pain	SE was converted to SD
	[D] Roland-Morris Disability Questionnaire	- Data were extracted from images - SD was imputed from mean change SD of Peloso et al., 2004.[b]
Skljarevski 2010[54]	[S] Athens Insomnia Scale [P] Brief Pain Inventory-average pain	SE was converted to SD
	[D] Roland-Morris Disability Questionnaire	SD was imputed from mean change SD of Peloso et al., 2004.[b]

Note: [S]=sleep; [P]=pain intensity; [D]=back-related disability

[a] Higgins JP, Li T, Deeks JJ. Chapter 6: Choosing effect measures and computing estimates of effect. 2021. In: Cochrane Handbook for Systematic Reviews of Interventions [Internet]. Cochrane. Available from: https://training.cochrane.org/handbook/current/chapter-06.

[b] Peloso PM, Fortin L, Beaulieu A, Kamin M, Rosenthal NR. Analgesic efficacy and safety of tramadol/ acetaminophen combination tablets (Ultracet) in treatment of chronic low back pain: A multicenter, outpatient, randomized, double blind, placebo controlled trial. Journal of Rheumatology. 2004;31(12):2454-63.

Appendix S1. Details of interventions, rescue medications, funding sources, and conflicts of interest for included studies

Table S6. Details of interventions from included studies

Study	Type of medication	Medication	Washout period	Treatment duration *(incl titration)*	Dose titration	Dose adjustment	Dose	Frequency
Buynak 2010[60]	Opioid	Tapentadol	1wk	15wk	3wk	Flexible dose	100-250mg	Twice daily
	Opioid	Oxycodone		15wk	3wk	Flexible dose	20-50mg	Twice daily
Christoph 2017[57]	Opioid	Tapentadol	3-21 days	14wk	2wk	Fixed does after titration	200mg	Twice daily
	Opioid	Cebranopadol	3-21 days	14wk	2wk	Fixed does after titration	200/400/600mg	Once daily
Cloutier 2013[58]	Opioid	Oxycodone	2-7 days[a]	4wk	Weekly	Flexible does	10-40mg	Twice daily
Gordon 2010a[61]	Opioid	Buprenorphine patch	2-7 days[a]	4wk	Weekly	Flexible dose	5-20µg/h	7-day patch
Gordon 2010b[62]	Opioid	Buprenorphine patch	2-7 days[a]	4wk	Weekly	Flexible dose	10-40µg/h	6-to-8-day patch
Lee 2013[59]	Opioid	Tramadol	No	4wk	1wk	Flexible dose	75-150mg	Twice daily
Manning 2017[56]	IMiD	Lenalidomide	No	12wk	Not reported	Fixed dose	10mg	Once daily
Skljarevski 2009[55]	Antidepressant	Duloxetine	No	13wk	1wk	Fixed dose	20/60/120mg	Once daily
Skljarevski 2010[54]	Antidepressant	Duloxetine	No	13wk	1wk at Week 1 and again at Week 8	Fixed does	60/120mg	Once daily

Note: IMiD= immunomodulating drug.

[a] Crossover randomised controlled trial; This indicated a washout period before randomisation. There was no washout period between treatment periods.

Table S7. Details of rescue medications for breakthrough pain

Study	Rescue medication
Buynak 2010[60]	During the titration period, acetaminophen was permitted (≤1000 mg/day as needed) as rescue medication except for the last 3 days of this period, when stability of the optimal dose of study medication was to be achieved. During the maintenance period (after dose titration), additional analgesic medication was not allowed unless deemed necessary for reasons unrelated to low back pain; and then, only acetaminophen (≤1000 mg/day) was allowed, for a maximum of 3 consecutive days.
Christoph 2017[57]	Paracetamol/acetaminophen was allowed as a rescue medication for unacceptable chronic low back pain (maximum total daily dose of 2 g and maximum 20 days of use during the maintenance phase (after dose titration)
Cloutier 2013[58]	Acetaminophen plus codeine 300 mg/30 mg to be taken every 4 h to 6 h as required for rescue analgesia. Patients were also provided with senna/docusate sodium tablets to treat constipation as required.
Gordon 2010a[61]	Acetaminophen 300 mg/codeine 30 mg, 1-2 tablets every 4-6 hours as needed, for rescue analgesia.
Gordon 2010b[62]	Rescue analgesia was provided as acetaminophen 325 mg to be taken as 1-2 tablets every 4-6 hours as needed.
Lee 2013[59]	The NSAIDs or COX-2–selective inhibitors that patients had been taking since 1week before study drug administration were maintained at the same dose levels during the test period. Othe pain relievers (including antidepressants, anticonvulsants, corticosteroids, and narcotic analgesics), were not permitted during the test period.
Manning 2017[56]	Acetaminophen 325mg or short-acting opioid for <7 days to treat pain flares, trauma, or procedural pain
Skljarevski 2009[55]	Episodic use of short-acting analgesics (<3 consecutive days and <20 total days) for the management of breakthrough chronic low back pain (rescue therapy) or acute conditions unrelated to lower back.
Skljarevski 2010[54]	Episodic use of short-acting analgesics (<3 consecutive days and <20 total days) for the management of breakthrough chronic low back pain (rescue therapy) or acute conditions unrelated to lower back.

Table S8. Funding sources and conflicts of interest for included studies

Study	Funding sources	Conflict of interest
Buynak 2010[60]	The trial was funded by Johnson & Johnson Pharmaceutical Research & Development L.L.C., and Global Development, Grunenthal GmbH, Aachen, Germany	One or more author(s) was/were or is/are employee(s) or shareholder(s) of Johnson & Johnson Pharmaceutical Research & Development L.L.C., and Global Development, Grunenthal GmbH, Aachen, Germany (funders) R Buynak, who is not employed by Johnson & Johnson Pharmaceutical Services, L.L.C., or Grunenthal GmbH, was not compensated for this manuscript. Although writing assistance was provided, the authors retained full editorial control over the content of the manuscript.
Christoph 2017[57]	Grunenthal GmbH funded and designed the trial, and analyzed and interpreted the data.	One or more author(s) are employees of Grunenthal GmbH. At the time of the trial and/or received research support, personal consulting, or lecture fees from Astellas, Develco Pharma, Galapagos, Grunenthal, Lilly, Merck, Mitsubishi Tanabe Pharma, and Pfizer.
Cloutier 2013[58]	The research was supported by a research grant from Purdue Pharma.	Not reported.
Gordon 2010a[61]	The study was funded by a research grant from Purdue Pharma, Canada.	Not reported.
Gordon 2010b[62]	This study was funded by Purdue Pharma, Pickering, Ontario, Canada.	One or more author(s) was/were or is/are employee of Purdue Pharma (funder) or received compensation for their participation.
Lee 2013[59]	This research was financially supported by a grant from Janssen Korea, Ltd, whose role was restricted to providing assistance in the conception, conduct, data acquisition, and analysis of this study.	All authors have received research funding from Janssen Korea, Ltd.
Manning 2017[55]	The study was sponsored by Celgene Corporation.	Celgene Corporation (funder) developed the protocol in conjunction with investigators and was solely responsible for site selection, data collection, storage technology, clinical monitoring, statistical analysis, provision of study drug and matching control, and reporting of efficacy and safety data to regulatory agencies. One or more author(s) was/were or is/are employee of Celgene Corporation (funder) or the owner of the Celgene Corporation stock.
Skljarevski 2009[55]	Study design, funding and drugs were supplied by Eli Lilly and Company.	One or more author(s) was/were or is/are employees of Celgene Corporation (funder) or the shareholders of the Eli Lilly and Company (funder).
Skljarevski 2010[54]	Sponsored by Eli Lilly and Company, Indianapolis, USA.	One or more of the author(s) has/have received or will receive benefits for personal or professional use from a commercial party related directly or indirectly to the subject of this manuscript: e.g., honoraria, gifts, consultancies, royalties, stocks, stock options, decision-making position.

Table S9. List of adverse events reported by ≥10% of participants in each included study

Study	Medication type	Treatment group	Control group
Skljarevski 2010[54]	Antidepressant	No details reported	No details reported
Skljarevski 2009[55]	Antidepressant	No details reported	No details reported
Manning 2017[56]	IMiD	Constipation, rash, pruritus	Each adverse event was reported by <10% of participants
Christoph 2017[57]	Opioid	Dizziness, nausea, somnolence, vomiting, constipation, fatigue, headache	Each adverse event was reported by <10% of participants
Cloutier 2013[58]	Opioid	Nausea	Nausea
Lee 2013[59]	Opioid	Nausea, dizziness, constipation, vomiting, pruritus, somnolence, dyspepsia	Nausea, dyspepsia
Buynak 2010[60]	Opioid	Nausea, constipation, vomiting, headache, dizziness, somnolence, pruritus	Headache, dizziness
Gordon 2010a[61]	Opioid	Nausea, somnolence, pruritus, asthenia, constipation, insomnia, dizziness, sweating, pain, anorexia, vomiting, yawn, nervousness, rash, headache, diarrhea	Pruritus, pain, insomnia, constipation, nervousness, diarrhea, anorexia, yawn, nausea, asthenia, sweating, rash
Gordon 2010b[62]	Opioid	Nausea, dizziness, pruritus, vomiting, somnolence, rash, dry mount, constipation, sweating, asthenia, headache	Nausea, pruritus, rash

Note: IMiD= immunomodulating drug.

Appendix S2. Risk of bias assessment

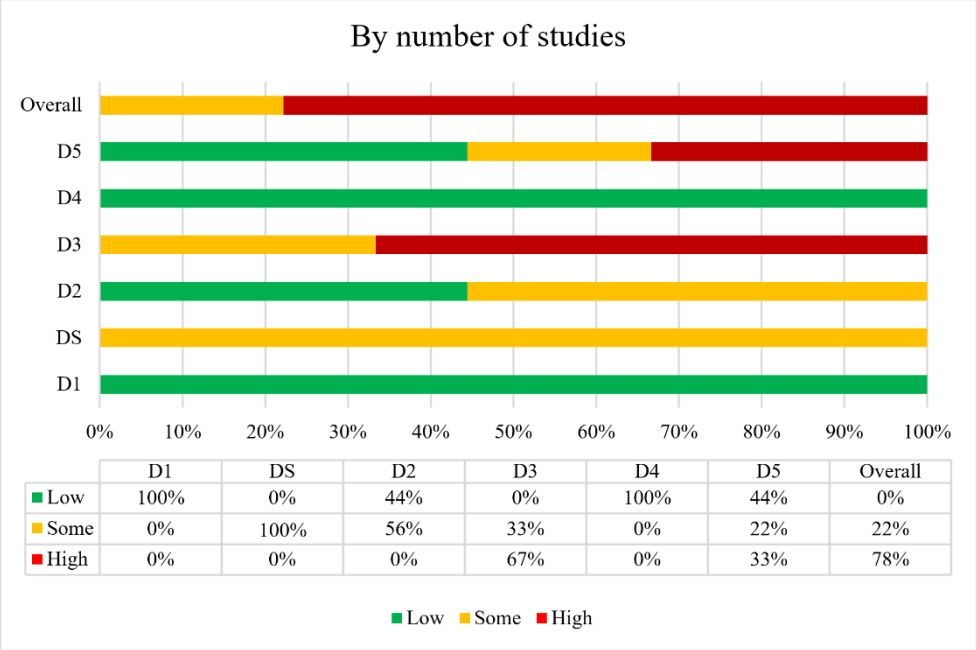
Figure S1. Risk of bias assessment for included studies

Author	Year	Medication type	Trial design	D1	DS	D2	D3	D4	D5	Overall	n randomised
Buynak	2010[60]	Opioid	Parallel	+		+	×	+	+	×	981
Christoph	2017[57]	Opioid	Parallel	+		?	×	+	+	×	641
Cloutier	2013[58]	Opioid	Crossover	+	?	?	?	+	?	?	83
Gordon	2010a[61]	Opioid	Crossover	+	?	?	?	+	+	?	79
Gordon	2010b[62]	Opioid	Crossover	+	?	?	?	+	×	×	78
Lee	2013[59]	Opioid	Parallel	+		?	×	+	×	×	245
Manning	2017[56]	Opioid	Parallel	+		+	×	+	?	×	180
Skljarevski	2009[55]	Opioid	Parallel	+		+	×	+	+	×	404
Skljarevski	2010[54]	IMiD	Parallel	+		+	×	+	×	×	236

+ Low ? Some × High

Note: IMiD= immunomodulating drug. D1=randomisation process; DS=risk of bias from period and crossover effects; D2=deviations from the intended interventions (assignment); D3=missing outcome data; D4=measurement of the outcome; D5=selection of the reported result; Overall=overall risk of bias for each study.

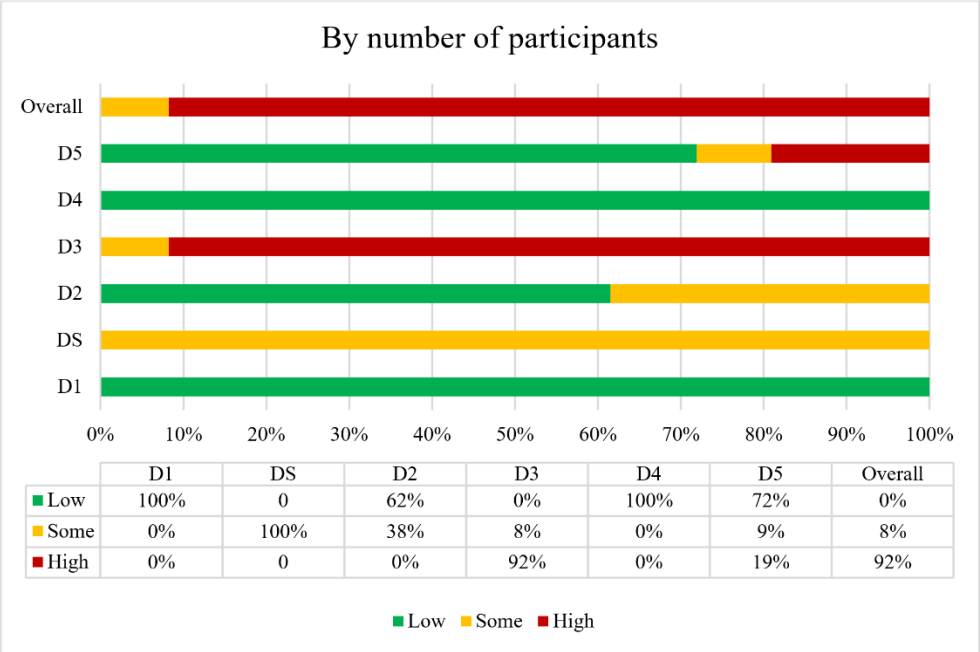
Figure S2. Risk of bias assessment for included studies (by number of studies)



	D1	DS	D2	D3	D4	D5	Overall
Low	100%	0%	44%	0%	100%	44%	0%
Some	0%	100%	56%	33%	0%	22%	22%
High	0%	0%	0%	67%	0%	33%	78%

Note: D1=randomisation process; DS=risk of bias from period and crossover effects; D2=deviations from the intended interventions (assignment); D3=missing outcome data; D4=measurement of the outcome; D5=selection of the reported result; Overall=overall risk of bias for each study.

Figure S3. Risk of bias assessment for included studies (by number of participants)

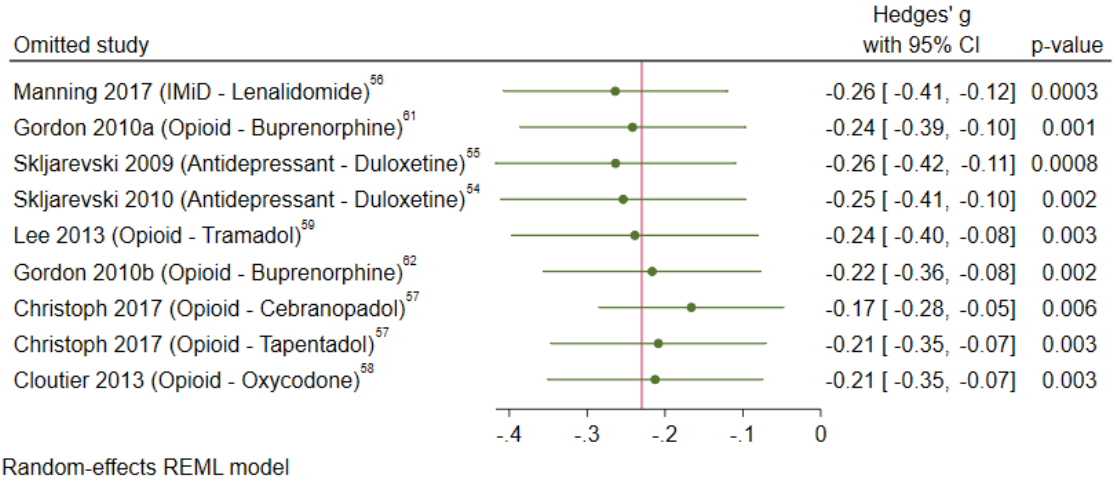


	D1	DS	D2	D3	D4	D5	Overall
Low	100%	0	62%	0%	100%	72%	0%
Some	0%	100%	38%	8%	0%	9%	8%
High	0%	0	0%	92%	0%	19%	92%

Note: D1=randomisation process; DS=risk of bias from period and crossover effects; D2=deviations from the intended interventions (assignment); D3=missing outcome data; D4=measurement of the outcome; D5=selection of the reported result; Overall=overall risk of bias for each study.

Appendix S3. Sensitivity analysis (leave-one-out) for sleep outcomes

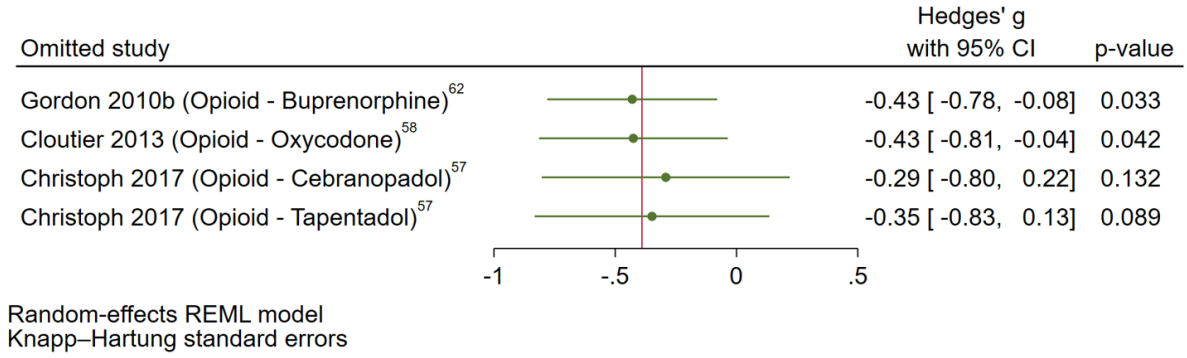
Figure S4. Sensitivity analysis (leave-one-out) for sleep



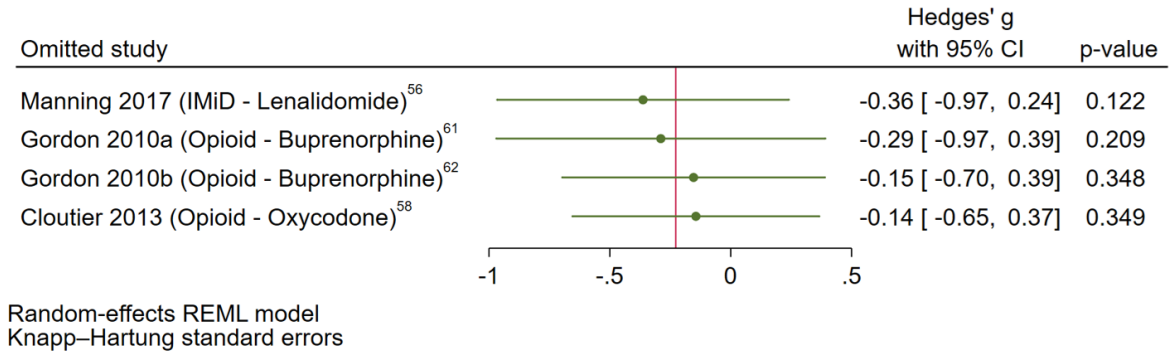
Note: IMiD= immunomodulating drug

Figure S5. Sensitivity analysis (leave-one-out) for different sleep characteristics

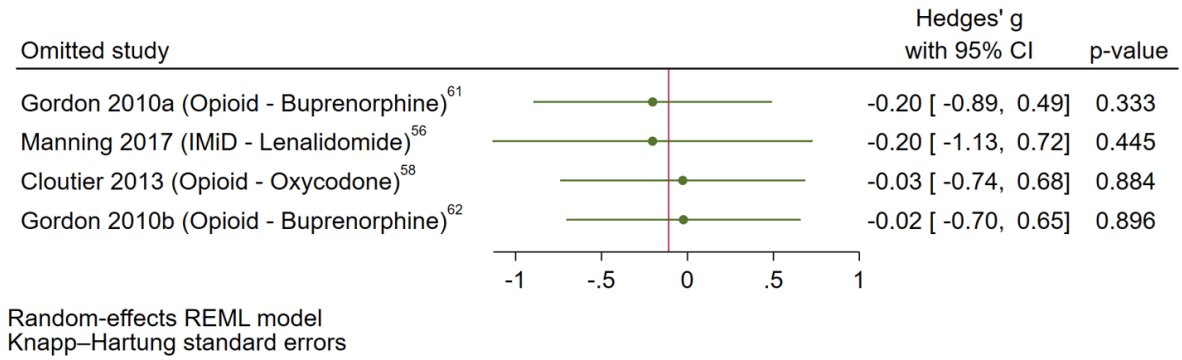
a) Sleep Quality



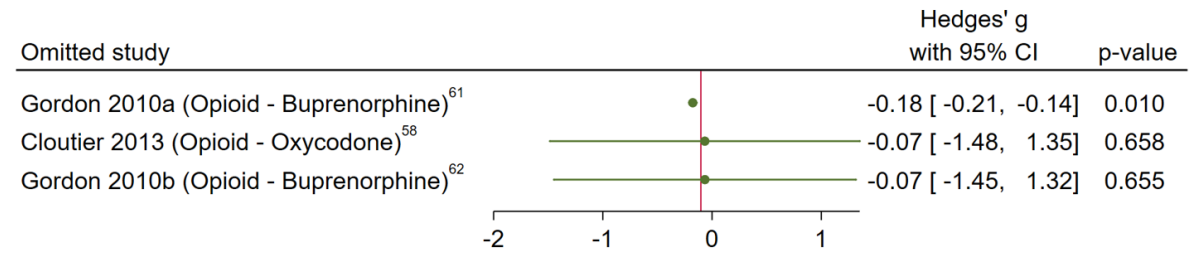
b) Trouble Falling Asleep



c) Trouble Staying Asleep/Wake After Sleep Onset



d) Medication Use to Aid Sleep



Note: IMiD= immunomodulating drug. The Hartung-Knapp-Sidik-Jonkman method was used as there were ≤5 studies in each analysis.

Figure S6. Funnel plot for sleep

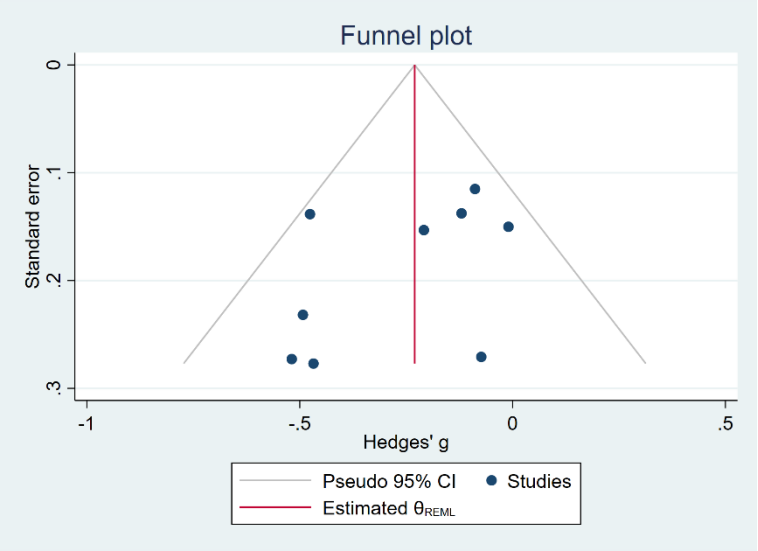
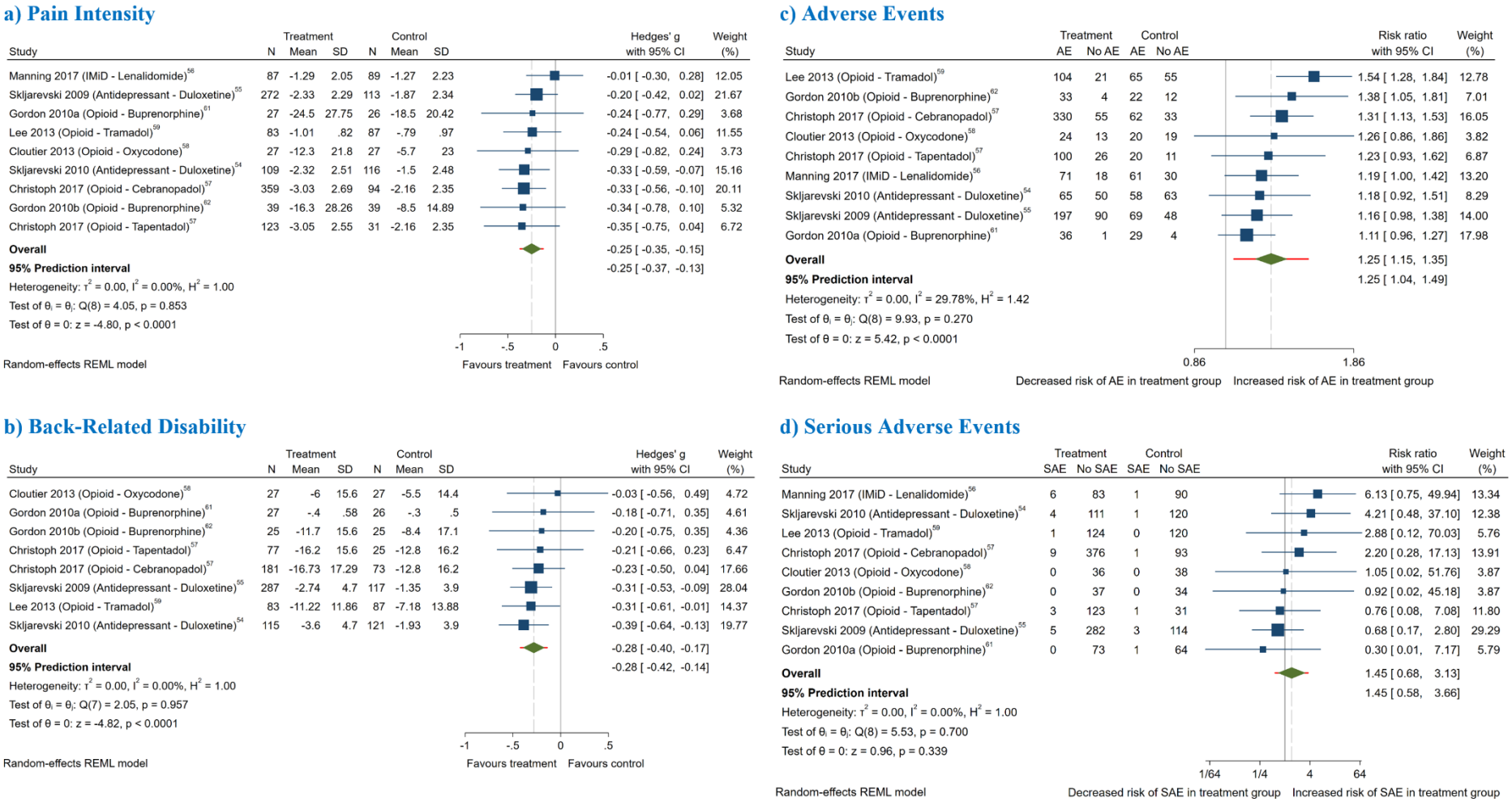


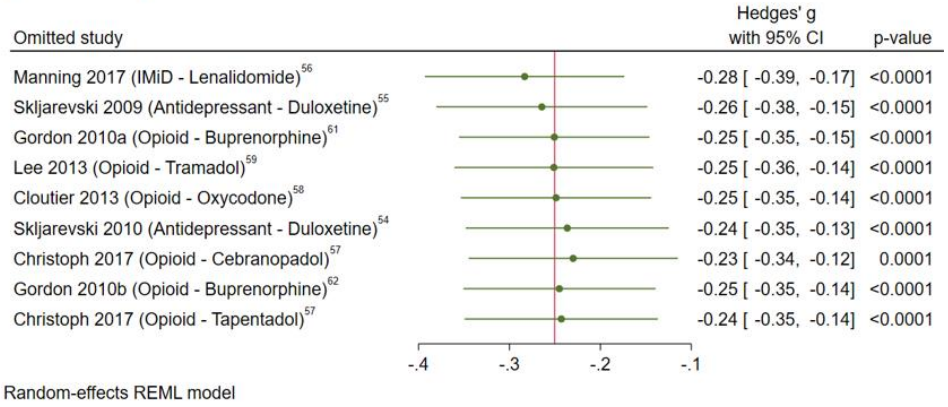
Figure S7. Effect of pharmacotherapy on secondary outcomes in adults with chronic low back pain.



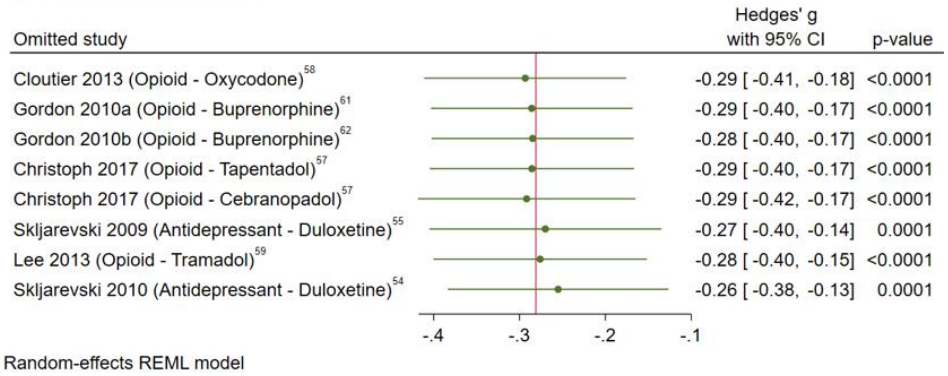
Note: Negative g-values indicated the result favoured the experimental group (i.e., pharmacotherapy resulted in better health outcomes). Risk ratio value >1 indicated higher risk of adverse events or serious adverse events in the treatment group. AE = adverse events; IMiD= immunomodulating drug; SAE = serious adverse events.

Figure S8. Sensitivity analysis (leave-one-out) for secondary outcomes

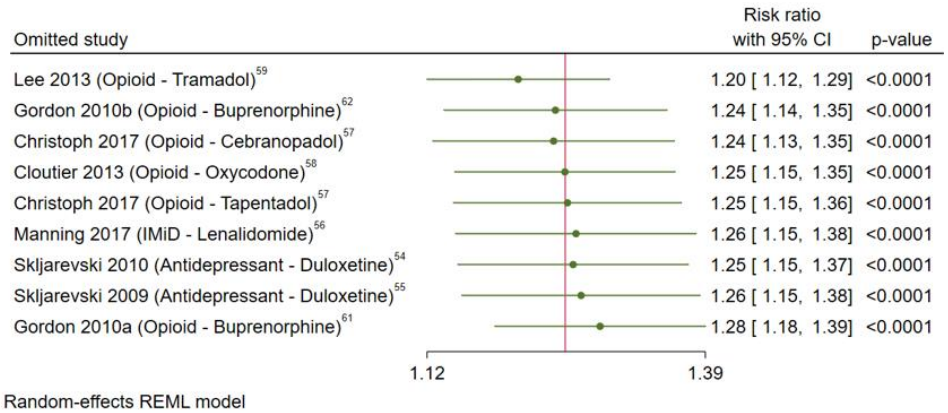
a) Pain Intensity



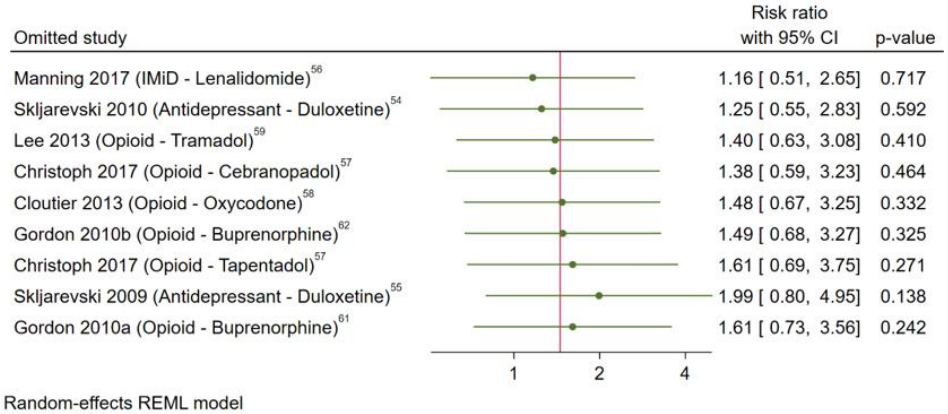
b) Back-Related Disability



c) Adverse Events



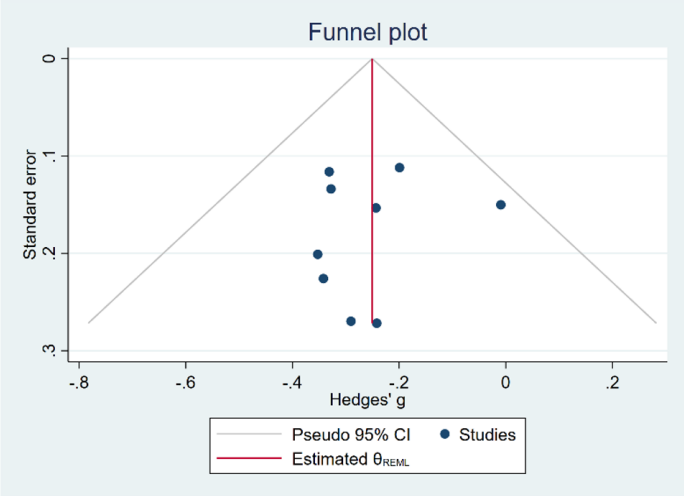
d) Serious Adverse Events



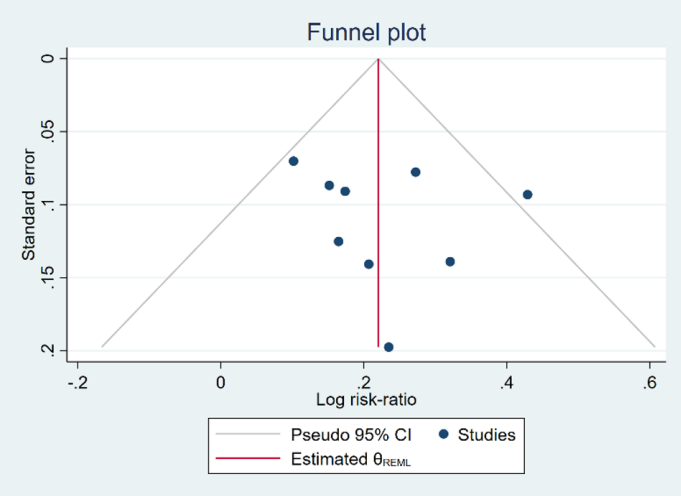
Note: IMiD= immunomodulating drug.

Figure S9. Funnel plots for secondary outcomes

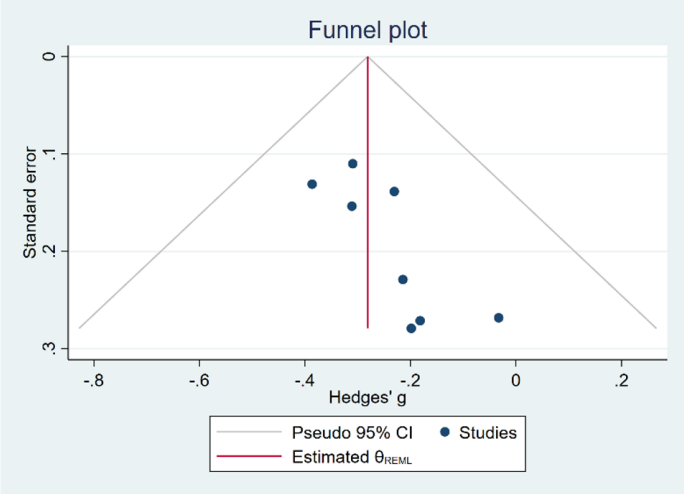
a) Pain Intensity



c) Adverse Events



b) Back-Related Disability



d) Serious Adverse Events

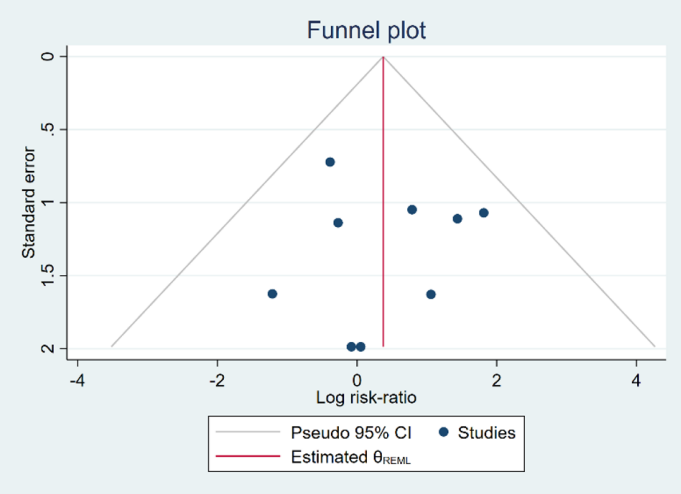


Table S10. Subgroup analysis (opioids)

Outcome	Studies	*n* analysed	Effect estimates	*P*-value	I^2 (%)	Prediction interval[a]	Compared to effect estimates of overall pharmacotherapy[b]
			Hedges' *g* [95% CI]				
Sleep	5	686	**-0·37 [-0·53, -0·21]**	**<·0001**	0·0%	-0·60, -0·15	No change
Sleep quality	3	463	**-0·39 [-0·66, -0·12]**	**·020**	0·0%	NP	No change
Insomnia symptom severity	NR						
Total sleep time	2	106	0·20 [-0·98, 1·37]	·280	0·0%	NP	No change
Trouble falling asleep	3	159	-0·36 [-0·97, 0·24]	·120	0·0%	NP	No change
Trouble staying sleep/WASO	3	159	-0·20 [-1·13, 0·72]	·450	46·5%	NP	No change
Use of medication to aid sleep	3	159	-0·10 [-0·42, 0·21]	·300	0·0%	NP	No change
Pain intensity	5	686	**-0·31 [-0·45, -0·16]**	**<·0001**	0·0%	-0·51, -0·10	No change
Back-related disability	5	686	**-0·23 [-0·38, -0·07]**	**·010**	0·0%	-0·45, -0·002	No change
			Risk ratio [95% CI]				
Adverse events	5	686	**1·29 [1·15, 1·45]**	**<·0001**	45·3%	0·95, 1·76	No change
Serious adverse events	5	686	1·16 [0·37, 3·64]	·800	0·0%	0·23, 5·85	No change

Note: Bold texts indicate significant results. Negative *g*-values indicated the result favoured the experimental group (i.e., pharmacotherapy resulted in better health outcomes), except for 'total sleep time' where negative g-values indicated pharmacotherapy resulted in less total sleep time. Risk ratio value >1 indicated higher risk of adverse events or serious adverse events in the treatment group. NP=Not performed; NR=not reported; WASO=wake after sleep onset.

[a] Prediction interval indicates the range of potential values that could be possible for future trials.

[b] Results are considered changed if g-value changed between effect and no effect, or if *g*-value changed ≥0·3.

Appendix S4. Dataset and statistical codes for meta-analyses

Table S11. Dataset for meta-analyses

study_id	sleep_exp_n	sleep_con_n	sleep_exp_chg	sleep_exp_chg_sd	sleep_con_chg	sleep_con_chg_sd	qual_exp_n	qual_con_n	qual_exp_eot	qual_exp_eot_sd	qual_con_eot	qual_con_eot_sd
Christoph 2017 (Cebranopadol)	182	75	-70	27	-57	29	182	75	-70	27	-57	29
Christoph 2017 (Tapentadol)	75	25	-70	26	-57	29	75	25	-70	26	-57	29
Cloutier 2013	27	27	-85	118	-28	99	27	27	-38	27	-34	23
Gordon 2010a	27	26	-72	82	-66	73						
Gordon 2010b	26	26	-106	143	-51	81	26	26	-40	24	-36	26
Lee 2013	83	87	-1	1	0	1						
Manning 2017	87	89	-1	2	-1	2						
Skljarevski 2009	254	107	-2	4	-1	4						
Skljarevski 2010	104	106	-2	5	-1	5						

study_id	hrs_exp_n	hrs_con_n	hrs_exp_eot	hrs_exp_eot_sd	hrs_con_eot	hrs_con_eot_sd	falling_exp_n	falling_con_n	falling_exp_eot	falling_exp_eot_sd	falling_con_eot	falling_con_eot_sd
Christoph 2017 (Cebranopadol)												
Christoph 2017 (Tapentadol)												
Cloutier 2013	27	27	6	2	6	2	27	27	41	29	56	31
Gordon 2010a							27	26	36	29	38	30
Gordon 2010b	26	26	6	2	6	1	26	26	36	29	51	30
Lee 2013												
Manning 2017							87	89	-1	2	-1	2
Skljarevski 2009												
Skljarevski 2010												

study_id	staying_exp_n	staying_con_n	staying_exp_eot	staying_exp_eot_sd	staying_con_eot	staying_con_eot_sd	med_exp_n	med_con_n	med_exp_eot	med_exp_eot_sd	med_con_eot	med_con_eot_sd
Christoph 2017 (Cebranopadol)												
Christoph 2017 (Tapentadol)												
Cloutier 2013	27	27	45	33	58	31	27	27	34	37	41	38
Gordon 2010a	27	26	42	32	35	32	27	26	27	35	26	36
Gordon 2010b	26	26	41	32	55	30	26	26	29	32	36	35
Lee 2013												
Manning 2017	87	89	-1	2	-1	2						
Skljarevski 2009												
Skljarevski 2010												

study_id	pain_exp_n	pain_con_n	pain_exp_chg	pain_exp_chg_sd	pain_con_chg	pain_con_chg_sd	dis_exp_n	dis_con_n	dis_exp_chg	dis_exp_chg_sd	dis_con_chg	dis_con_chg_sd
Christoph 2017 (Cebranopadol)	359	94	-3	3	-2	2	181	73	-17	17	-13	16
Christoph 2017 (Tapentadol)	123	31	-3	3	-2	2	77	25	-16	16	-13	16
Cloutier 2013	27	27	-12	22	-6	23	27	27	-6	16	-6	14
Gordon 2010a	27	26	-25	28	-19	20	27	26	0	1	0	1
Gordon 2010b	39	39	-16	28	-9	15	25	25	-12	16	-8	17
Lee 2013	83	87	-1	1	-1	1	83	87	-11	12	-7	14
Manning 2017	87	89	-1	2	-1	2						
Skljarevski 2009	272	113	-2	2	-2	2	287	117	-3	5	-1	4
Skljarevski 2010	109	116	-2	3	-2	2	115	121	-4	5	-2	4

study_id	ae_exp_n_analysed	ae_exp_y	ae_exp_none	ae_con_n_analysed	ae_con_y	ae_con_none	sae_exp_y	sae_exp_none	sae_con_y	sae_con_none
Christoph 2017 (Cebranopadol)	385	330	55	95	62	33	9	376	1	93
Christoph 2017 (Tapentadol)	126	100	26	31	20	11	3	123	1	31
Cloutier 2013	37	24	13	39	20	19	0	36	0	38
Gordon 2010a	37	36	1	33	29	4	0	73	1	64
Gordon 2010b	37	33	4	34	22	12	0	37	0	34
Lee 2013	125	104	21	120	65	55	1	124	0	120
Manning 2017	89	71	18	91	61	30	6	83	1	90
Skljarevski 2009	287	197	90	117	69	48	5	282	3	114
Skljarevski 2010	115	65	50	121	58	63	4	111	1	120

Table S12. Stata statistical codes for meta-analyses

Outcome	Statistical code
Overall sleep impairments	. meta esize sleep_exp_n sleep_exp_chg sleep_exp_chg_sd sleep_con_n sleep_con_chg sleep_con_chg_sd, studylabel(study_id) eslabel(Hedges' g) gsort -_meta_es meta summarize, random(reml) predinterval(95)
Sleep quality	. meta esize qual_exp_n qual_exp_eot qual_exp_eot_sd qual_con_n qual_con_eot qual_con_eot_sd, studylabel(study_id) eslabel(Hedges' g) gsort -_meta_es meta summarize, random(reml) se(khartung)
Insomnia symptom severity	. meta esize sleep_exp_n sleep_exp_chg sleep_exp_chg_sd sleep_con_n sleep_con_chg sleep_con_chg_sd, studylabel(study_id) eslabel(Hedges' g) gsort -_meta_es meta summarize, random(reml) se(khartung)
Total sleep time	. meta esize hrs_exp_n hrs_exp_eot hrs_exp_eot_sd hrs_con_n hrs_con_eot hrs_con_eot_sd, studylabel(study_id) eslabel(Hedges' g) gsort +_meta_es meta summarize, random(reml) se(khartung)
Trouble falling asleep	. meta esize falling_exp_n falling_exp_eot falling_exp_eot_sd falling_con_n falling_con_eot falling_con_eot_sd, studylabel(study_id) eslabel(Hedges' g) gsort -_meta_es meta summarize, random(reml) se(khartung)
Trouble staying asleep/wake after sleep onset	. meta esize staying_exp_n staying_exp_eot staying_exp_eot_sd staying_con_n staying_con_eot staying_con_eot_sd, studylabel(study_id) eslabel(Hedges' g) gsort -_meta_es meta summarize, random(reml) se(khartung)
Use of medication to aid sleep	. meta esize med_exp_n med_exp_eot med_exp_eot_sd med_con_n med_con_eot med_con_eot_sd, studylabel(study_id) eslabel(Hedges' g) gsort -_meta_es meta summarize, random(reml) se(khartung)
Pain intensity	. meta esize pain_exp_n pain_exp_chg pain_exp_chg_sd pain_con_n pain_con_chg pain_con_chg_sd, studylabel(study_id) eslabel(Hedges' g) gsort -_meta_es meta summarize, random(reml) predinterval(95)
Back-related disability	. meta esize dis_exp_n dis_exp_chg dis_exp_chg_sd dis_con_n dis_con_chg dis_con_chg_sd, studylabel(study_id) eslabel(Hedges' g) gsort -_meta_es meta summarize, random(reml) predinterval(95)
Adverse events	. meta esize ae_exp_y ae_exp_none ae_con_y ae_con_none, esize(lnrratio) studylabel(study_id) gsort -_meta_es meta summarize, random(reml) predinterval(95) meta summarize, random(reml) transform(exp) predinterval(95)
Serious adverse events	. meta esize sae_exp_y sae_exp_none sae_con_y sae_con_none, esize(lnrratio) studylabel(study_id) gsort -_meta_es meta summarize, random(reml) predinterval(95) meta summarize, random(reml) transform(exp) predinterval(95)